AI101 MINI-PROJECT | TEAM NO. 44

Movie Recommender

A genre-based movie recommender using Logistic Regression and Jaccard similarity on Netflix titles.

GITHUB REPO

https://github.com/2026eeb1309/AI101-Mini-project.git

MEMBERS & ROLES

1 **Aadarsh Kumar**
Entry No: 2026EEB1309
EXECUTION LEAD

2 **Ankush Chaudhary**
Entry No: 2026AIB1017
DOCUMENTATION LEAD

3 **Aryan Rahul Bachchhav**
Entry No: 2026MCB1517
VIDEO LEAD

4 **Abhinav Mishra**
Entry No: 2026DAB1289
PPT LEAD

Problem & Data

What we are solving, why it matters, and the dataset behind it

THE PROBLEM: WHAT AND WHY

What: given one movie a user likes, suggest 5 other movies they may enjoy.

Why: Netflix has thousands of titles, so finding similar ones by scrolling is slow. Genre is the clearest signal in the data, so we use it to match movies.

Task: Logistic Regression learns Movie vs TV Show from genres (binary classification); Jaccard genre similarity then ranks the closest movies.

DATASET

SOURCE

Kaggle: Netflix titles (netflix_titles.csv)
kaggle.com/code/abhichoudhury/a-simple-movie-recommender

SIZE

6,234 titles: 4,265 Movies and 1,969 TV Shows, released 1925 to 2020.

COLUMNS

12 columns: show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, description.

FIRST ROWS OF THE DATASET

netflix_titles.csv — 6,234 rows × 12 columns

show_id	type	title	release_year	duration	listed_in
81145628	Movie	Norm of the North: Ki...	2019	90 min	Children & Family Movies, Com...
80117401	Movie	Jandino: Whatever it ...	2016	94 min	Stand-Up Comedy
70234439	TV Show	Transformers Prime	2013	1 Season	Kids' TV
80058654	TV Show	Transformers: Robots ...	2016	1 Season	Kids' TV
80125979	Movie	#realityhigh	2017	99 min	Comedies
80163890	TV Show	Apaches	2016	1 Season	Crime TV Shows, International...
70304989	Movie	Automata	2014	110 min	International Movies, Sci-Fi ...
80164077	Movie	Fabrizio Copano: Solo...	2017	60 min	Stand-Up Comedy

First 8 rows shown. Other columns: director, cast, country, date_added, rating, description.

6,234	12	1925-2020
TITLES	COLUMNS	RELEASE YEARS

Approach

Pipeline: data → model → output, and the libraries we used

1 DATA

- Netflix titles dataset from Kaggle: 6,234 rows.
- Clean it: remove duplicates, drop missing values, fix types.
- Turn genres into 0/1 columns with MultiLabelBinarizer: 42 genres + release_year = 43 features.

2 MODEL

- Split 80% train / 20% test (stratified).
- SelectKBest (chi-square) keeps the best 15 genres.
- StandardScaler, then Logistic Regression (balanced classes, max_iter = 1000).

3 OUTPUT

- Movie / TV Show label with a probability: 99.04% accuracy.
- Jaccard genre similarity, movies only.
- Top 5 similar movies for the chosen movie.

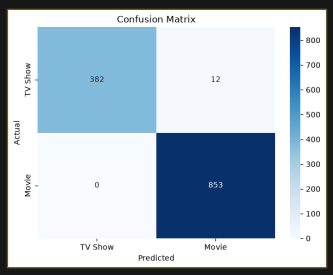
LIBRARIES USED

pandas	numpy	matplotlib	seaborn	scikit-learn
load and clean the data	numbers and RMSE	charts and curves	heatmaps and bar plots	split, select, scale, model, metrics

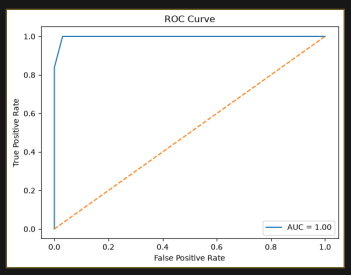
Results

Key charts, key numbers and a demo of the recommender

CONFUSION MATRIX



ROC CURVE



DEMO: recommend_movies("The Conjuring", 5)

```
Selected Movie: The Conjuring
Genres: ['Horror Movies', 'Thrillers']
========== RECOMMENDED MOVIES ==========
title            year  listed_in                similarity
Before I Wake    2016  Horror Movies, Thrillers     1.0
The Detained     2017  Horror Movies, Thrillers     1.0
Gothika          2003  Horror Movies, Thrillers     1.0
Clinical         2017  Horror Movies, Thrillers     1.0
Cabin Fever      2016  Horror Movies, Thrillers     1.0
```

KEY NUMBERS (TEST SET)

99.04% Accuracy	**98.61%** Precision
100% Recall	**99.30%** F1 score
0.9975 ROC AUC	**0.096** RMSE

Only 12 of 1,247 test titles were misclassified, and no Movie was missed.

Learnings

What worked, what didn't, who did what, and our AI-use disclosure

WHAT WORKED

- Genres alone separate Movies from TV Shows: 99.04% accuracy, AUC 0.9975.
- Fitting feature selection and scaling on training data only avoided data leakage.
- Balanced class weights handled 4,265 Movies vs 1,969 TV Shows.
- Jaccard similarity gave sensible picks (The Conjuring → Gothika, Cabin Fever).

WHAT DIDN'T

- Logistic Regression only predicts Movie vs TV Show; the recommendations come from the Jaccard step.
- Many movies tie at similarity 1.0, so the top 5 are not truly ranked.
- Only genres are used (no ratings or user history), so it is not personalised.
- release_year did not make the top 15 features, and the title must be typed in full.

WHO DID WHAT

Aadarsh Kumar EXECUTION LEAD	Code execution and results
Ankush Chaudhary DOCUMENTATION LEAD	README and documentation
Aryan Rahul Bachchhav VIDEO LEAD	Project demo video
Abhinav Mishra PPT LEAD	This presentation

AI-USE DISCLOSURE

Some parts of the notebook were written with the help of AI, as they were beyond our current level. The statistics part and the Logistic Regression part were written by us. This PPT was made with AI, but the prompt for the theme and content came from Group-44.